Solutions

Question 1 on Wind and Photovoltaic Power

a)

- The efficiency of a wind turbine depends on the tip-speed ratio $\lambda = \frac{\omega_T R}{v_w}$ and is maximum for a particular $\lambda = \lambda_{\text{opt}}$
- For different wind speeds, the rotational speed of the wind turbine needs to vary according to $\omega_T = \frac{\lambda_{opt} \times v_w}{R}$ to keep operating at the optimum tip-speed ratio and thus, maximise the power output.
- Apart from maximising power capture, variable speed operation reduces the mechanical stress on the wind turbines wind speed varies.

[1 mark each – 3 marks]

b)

- If the force applied on the wind turbine is too high, the wind speed drops too far, and the power extracted which is the product of force and wind speed at the plane of rotation of the turbine falls.
- On the other hand a smaller force will allow wind to pass over the blades at higher speed but will not extract much power. Hence, there is an optimum force that maximises the power extracted.

[1 mark each – 2 marks]

c)

i) Rated wind speed

$$P_w = \frac{1}{2}\rho A v_w^3 C_{p-max}$$

$$\Rightarrow v_w = \left(\frac{2P_{rated}}{\rho \pi r^2 C_{p-max}}\right)^{\frac{1}{3}} = 10.46 \text{ m/s}$$

[1 mark]

$$p = \Phi(10.46) - \Phi(3) = 68.68\%$$

[1 mark]

iv)

$$\text{TSR } \lambda = \frac{\omega_T \times r}{v_w} \Rightarrow \omega_{T-opt} = \frac{\lambda_{opt} \times v_w}{r}$$

Model Answers

[1 mark]

$$\left(\frac{\lambda_{opt} \times v_{w-cutin}}{r} \times \frac{60}{2\pi}\right) \leq \omega_{T-opt} \leq \left(\frac{\lambda_{opt} \times v_R}{r} \times \frac{60}{2\pi}\right)$$
$\Rightarrow 2.47$ rpm $\leq \omega_{T-opt} \leq 8.62$ rpm

[1 mark]

v)

Synchronous speed (in rpm)

$$N_s = \frac{60 \times f}{p/_2} \Rightarrow p = \frac{120 \times f}{N_s}$$
[0.5 mark]

For
$f \geq 10$ Hz, $p \geq \frac{120 \times f}{N_{s-min}} = 363.96$

Number of poles 364
[0.5 mark]

d)

i)

At solar noon
$\alpha = 90° - (\phi - \delta)$

[1 mark]

Irradiance on PV module
$I_{\text{module}} = I_{\text{incident}} \sin(\alpha + \beta)$

To maximise irradiance on PV module
$\alpha + \beta = 90° \rightarrow \beta = \phi - \delta = 38.5°$

[1 mark]

ii)

Area of each module

$$A = \frac{P_{\text{rated}}}{1000 \times \eta} = 1.83 \text{ m}^2$$

[1 mark]

Model Answers

$P_{\text{arr}} = N_{\text{mod}} \times 600 \times A \times \eta = 5.94$ kW

[1 mark]

iii)

The current fed by one affected string with two shaded modules would be zero. Hence power output will be

$$P_{\text{out}} = P_{\text{arr}} \times \frac{4}{5} = 4.75 \text{ kW}$$

[1 mark]

e)

i)

$$0 = 5.5 - I_0 \left[e^{\frac{qV_{oc}}{kT}} - 1 \right]$$

$$\Rightarrow V_{oc} = \frac{kT}{q} \ln\left(\frac{4.1 + I_0}{I_0}\right)$$

[1 mark]

$V_{oc} = 0.7977$ V

[1 mark]

ii)

$$I_{SC} = 5.5 - I_0 \left[e^{\frac{q \times 0}{kT}} - 1 \right] = 5.5 \text{ A}$$

[1 mark]

iii) $R = 0.3 = \frac{V}{I}$

$V = 5.5 \times 0.3 = 1.65$ V

$P = V \times I = 9.075$ W

[1 mark]

iv)

$$R = 2 = \frac{V}{I}$$

$I \approx V_{oc} \times 2 = 1.595$ A

$P \approx V_{oc} \times I = 1.27$ W

Less power would be drawn by the 2 Ω resistor

[1 mark]

Model Answers

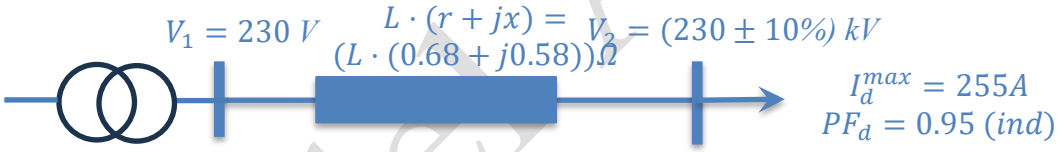
Question 2 on balancing, Voltage Control and Security of Supply

Part A

a)

Tap changer: passive voltage control by setting tap changer to compensate for voltage drop/rise along upstream feeder and transformer.
[0.5 marks]
On-load tap changer: active voltage control that compensates for voltage drop/rise along downstream feeders typically in accordance with downstream loading.
[0.5 marks]
In-line voltage regulators: located mid long feeder to increase/decrease voltage downstream voltage.
[0.5 marks]
Reactive power compensation: capacitors to increase voltage and reactors to decrease voltage at point of connection.
[0.5 marks]
[0.5 marks for any other explained approach for voltage control]
[Total number of marks is limited to 2]

b)



Sketch of the considered system

[1 mark]

$$V_2 = V_1 - \frac{L \cdot (r \cdot P_d + x \cdot Q_d)}{V_2}$$

$$Q_d = \sqrt{S_d^2 - P_d^2} = \sqrt{\frac{P_d^2}{PF_d^2} - P_d^2} = P_d \sqrt{\frac{1}{PF_d^2} - 1}$$

$$P_d^{max} = V_2^{min} \cdot I_d^{max} \cdot PF_d$$

$$L^{max} = \frac{\left(V_1 - V_2^{min}\right)}{I_d^{max} \cdot PF_d \cdot \left(r + x \cdot \sqrt{\frac{1}{PF_d^2} - 1}\right)}$$

[2 marks]

Model Answers

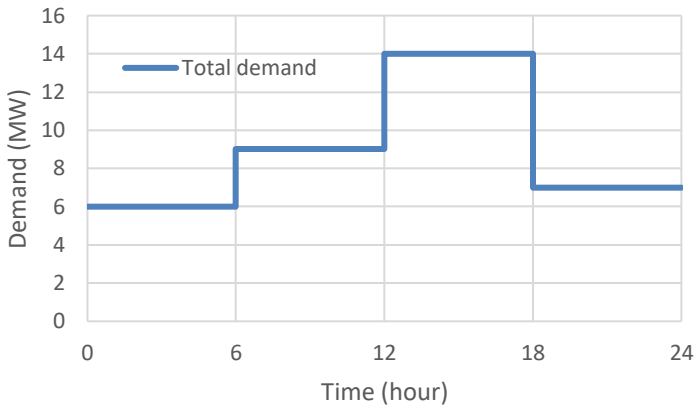
$$L_{\square}^{max} = \frac{(230 - 0.9 \cdot 230)}{255 \cdot 0.95 \cdot \left(0.68 + 0.58\sqrt{\frac{1}{0.95^2} - 1}\right)} = 0.1278km = 127.8m$$

[1 mark]

Part B

a)

Total demand profile is sketched below. Peak demand is 14 MW which is class of supply C.



[1 mark for correct peak demand value and specified class of supply; there is no need to sketch the total demand profile to get the full mark]

b)

Capacity outage probability table for the generation plant is

Capacity in service, MW	Probability
8	0.9 x 0.8 x 0.6 = 0.432
6	0.9 x 0.8 x (1-0.6) + 0.9 x (1-0.8) x 0.6 = 0.396
4	0.9 x (1-0.8) x (1-0.6) + (1-0.9) x 0.8 x 0.6 = 0.12
2	(1-0.9) x 0.8 x (1-0.6) + (1-0.9) x (1-0.8) x 0.6 = 0.044
0	(1-0.9) x (1-0.8) x (1-0.6) = 0.008

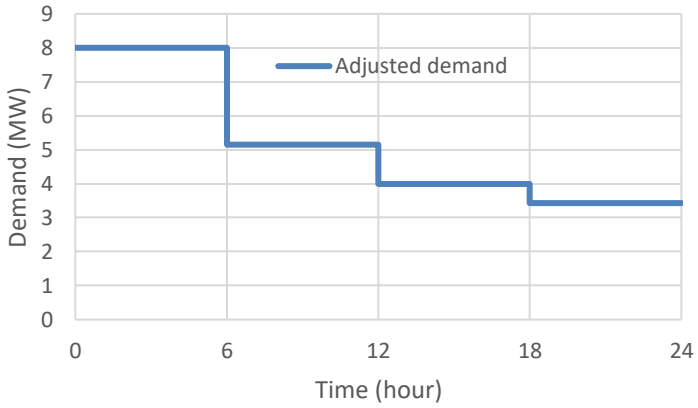
The total probability for all eight possible capacity states is 1.

[3 marks; subtract 1 mark for any missing capacity in service and wrong probability up to a maximum of subtracted 3 marks]

Model answers

c)

Total generation capacity is 8 MW and the total demand profile should be scaled to 8 MW keeping the same shape. Load duration curve is shown below with demand values 8, 5.1429, 4 and 3.4286 MW.



[0.5 marks for correct peak demand, 0.5 marks for correct shape]

d)

Energy not supplied for system state i, ENS_i, is calculated from the profile in the following way

$$ENS_i = \begin{cases} (8 - C_i) \cdot 6 & 5.1429 \leq C_i \leq 8 \\ (8 - C_i) \cdot 6 + (5.1429 - C_i) \cdot 6 & 4 \leq C_i \leq 5.1429 \\ (8 - C_i) \cdot 6 + (5.1429 - C_i) \cdot 6 + (4 - C_i) \cdot 6 & 3.4286 \leq C_i \leq 4 \\ (8 - C_i) \cdot 6 + (5.1429 - C_i) \cdot 6 + (4 - C_i) \cdot 6 + (3.4286 - C_i) \cdot 6 & 0 \leq C_i \leq 3.4286 \end{cases}$$

Where C_i is system state capacity in.

Expected energy not supplied is equal to $ENS \cdot Probability$

Capacity, MW	Probability	ENS, MWh	EENS, MWh
8	43.2%	0	0
6	39.6%	12	4.752
4	12%	30.857	3.7029
2	4.4%	75.429	3.3189
0	0.8%	123.43	0.98743
Total	**100.0%**		**12.761**

Total EENS is 12.761 MWh.

Model Answers

[4 marks; subtract 2 marks if it is not clearly explained how ENS and EENS are calculated; subtract 1 mark for any wrong pair ENS/EENS, including total EENS up to a maximum of subtracted 4 marks]

e)

Comparing total EENS and ENS from above COPT, the firm capacity is between 4 and 6 MW.

Energy not supplied when $C = 5.1429MW$ is

$$ENS_{5.1429} = (8 - 5.1429) \cdot 6 = 17.143MWh$$

The equivalent firm capacity is between 5.1429 and 6 MW.

[1 mark]

$$EENS = 12.761 = (8 - \mathrm{C}) \cdot 6$$

$$C = 8 - \frac{12.761}{6} = 5.8731MW$$

$$\frac{C}{8} = 0.73414 \approx 73.4\%$$

F Factor, of generating units, is 73.4%. Firm capacity is 5.8731 MW.

[1 mark; calculation of value of firm capacity in MW is sufficient for 1 mark]

f)

For class of supply C, two conditions, both within switching time, should be met:

- Within 15 minutes 2 MW or more, i.e. smaller of
  - Group Demand minus 12 MW; 14 – 12 = 2 MW
  - 2/3 Group Demand; 2/3 * 14 = 9.3 MW
- Within 3 hours Group Demand; 14 MW

Under first circuit outage, remaining capacity of one transformer is

$$7.5 \cdot 1.3 \cdot 1 = 9.75MW$$

[1 mark]

Therefore capacity to meet demand after FCO is

$$9.75 + 5.8731 = 15.623MW$$

[1 mark]

Model Answers

Given capacity to meet demand of 15.623 MW is greater than Group Demand of 14 MW, network is compliant with the security of supply standard at busbars 1.
[1 mark]

Model Answers

Question 3 on Low Carbon Heat

Part A

a)

Air source heat pump:

ASHP transfers thermal energy from the outside air to the area to be heated. There are 4 stages to and comprise:

- *Evaporation - The refrigerant passes through a heat exchanger where it evaporates absorbing heat in the process.*
- *Compression - The heated refrigerant is then compressed to higher pressure using an electrically driven compressor.*
- *Condensation - The compressed refrigerant passes through another heat exchanger located in the space to be heated and condenses, changing state back to a liquid and rejecting heat in the process.*
- *Expansion – The liquid refrigerant passes through an expansion device lowering its pressure prior to returning to the evaporator.*

[2 marks]

b)

District heating:

District heating is a system where heat for more than one building or an area is produced at a central location and distributed through a network of insulated pipes. Heat sources include:

- *Combined heat and power units*
- *Network heat pumps*
- *Waste(d) heat recovery*
- *Solar thermal*
- *Geothermal*

[any 4 of these would be sufficient for 1 mark]

The main components of a district heating system are:

- *Heat storage - to provide additional capacity and backup, etc.*
- *Heat network – connecting heat sources with buildings using supply and return pipes.*
- *Pumps – to circulate the hot water*
- *Heat substations – for pressure and temperature changes*
- *Heat metering – to monitor production and consumption*
- *Heat exchangers/ heat interface units - for connection of the heat network to buildings*

[any 4 of these would be sufficient for 1 mark]

c)

Hydrogen heating:

This is based on the repurposing of the existing gas distribution network from natural gas to hydrogen. Hydrogen can be produced using several processes but at present the most suited for large scale production is steam methane reforming using natural gas with carbon capture and storage and electrolysis using low carbon electricity.

[1 mark]

Model Answers

d)

Resistive heating:
There are a variety of different types of resistive heating systems. These include: radiant, convector, storage, direct electric, fan and panel heaters. However, they can only be regarded as low carbon if they use low carbon electricity.

[1 mark]

Part B

a) The purpose of the CCC is to:

- *advise UK Government and Devolved Administrations on emissions targets and report to Parliament on progress made in reducing greenhouse gas emissions and preparing for climate change*
- *provide independent advice on setting and meeting carbon budgets and preparing for climate change*
- *monitor progress in reducing emissions and achieving carbon budgets and targets*
- *conduct independent analysis into climate change science, economics and policy*
- *engage with a wide range of organisations and individuals to share evidence and analysis*

[any 3 of these would be sufficient for 1 mark]

b) The UK Government amended the target from 80% reduction of greenhouse gases by 2050 to 100% reduction to achieve Net Zero GHG emissions.

[1 mark]

Part C

a) Typical annual building space heat demand (MWh_{th}) = (Tint – T ext) * Average thermal transmittance * Building floor area*Nos of days heating is on*Nos of hours heating is on during the day

*= (18-7) *2.0*120*150*12 =* ***4,752,000Wh$_{th}$*** *or* ***4,752 kWh$_{th}$***

Where:
Tint = Building internal set temperature, °C
Text= Building external set temperature, °C

Assumptions:
Estimate of heat demand does not take account of:

- *other weather factors such as wind, solar gain.*

Model Answers

- *building air changes*
- *other heat sources*

[4 marks]

b)

i) Resistive heating cost:

*= Purchase and installation cost + Running costs*Space heat demand/ Efficiency*
*=£750pa + 0.3*4752/100% =* ***£2175.6 pa***

Air source heat pump costs:
*= Purchase and installation cost + Running costs*Space heat demand/ Efficiency*
*=£1500pa + 0.3*4752/300% =* ***£1975.2 pa***
The householder should select the ASHP system
[2 marks]

ii) Need to find the space heat demand, X, when: Cost of resistive heating = Cost of ASHP heating

*£750 + X * £0.3/100% = £1500 + X * £0.3 / 300%*

Rearranging: X = (£1500 - £750) / 0.2 = 3750 kWh pa

Hence space heat demand needs to be reduced from 4752 kWh to 3750 kWh which is: 100% - 3750/4752*100 = 21%
[3 marks]

iii) A heat pump can have an efficiency of greater than 100% as it excludes the energy absorbed from the air, i.e. it only includes the electrical energy for the compressor

[1 mark]

iv) A hybrid ASHP comprises a ASHP with a condensing boiler. This permits a smaller, lower cost heat pump to be used which may be a better match for the normal space heat demand of the building as it can operate at a higher efficiency. The condensing boiler operates when demand is high due to cold weather for example but it can also supply hot water demand if needed.

[2 marks]

Model Answers

Question 4 on Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

a) Vehicle-to-Grid (V2G) refers to the concept of discharging electricity from EV batteries and injecting it back into the grid in order to provide flexibility services and support the electricity system and/or the local network.

[1 marks for the explanation of V2G concept]

Key challenges include the cost of V2G charging hardware, battery degradation due to more frequent charging/discharging cycles and the lack of consumer awareness of the potential benefits.

[0.5 marks for the explanation of key challenges]

Key opportunities include the ability to provide a range of flexible system services, including frequency response, avoiding network reinforcement, supporting variable renewables, and reducing firm capacity requirements.

[0.5 marks for the explanation of key opportunities]

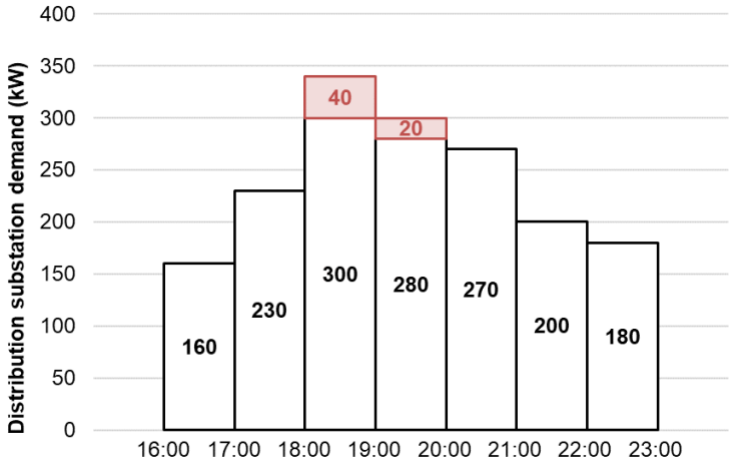
b)

i) The new substation peak demand will also occur at 6pm. It will be equal to the sum of baseline demand and the demand of 10 EVs in the first hour of their charging:

$$P_{\text{peak}} = 300\ \text{kW} + 10 \times 4\ \text{kW} = 340\ \text{kW}$$

[1 marks for the correct calculation of new peak demand]

The new demand diagram will be as shown in the chart below.



[1 marks for the correct demand chart]

ii) Given that the 10 EVs require 40 kW in the first hour of charging, they cannot all start charging together before 9pm without increasing the peak demand. In this case the EV charging would be completed by 11pm.

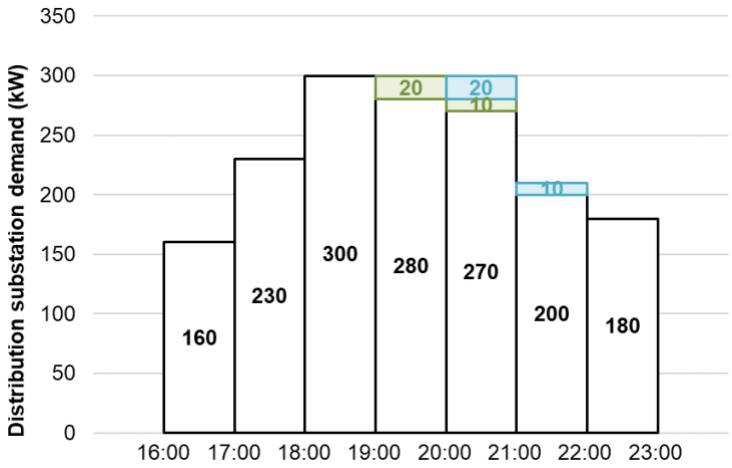
Model Answers

However, because EVs do not all have to follow the same charging schedule, it is better to charge half (5) of them from 7pm and then the other half (5) from 8pm without increasing the peak demand. With that approach the EV charging can be completed by 10pm without increasing the peak demand beyond the baseline 300 kW. Peak demand is now spread across three contiguous hours (6-9pm).

[2 marks for the correct time of the earliest completion of EV charging without increasing peak demand]

[2 marks for the correct EV charging schedule]

The new demand diagram is shown in the chart below, distinguishing between baseline demand and the charging demand for the two groups of EVs.



[2 marks for the correct demand chart]

Model answers

Hydro Power

a)

i)

$h = h_{hw} - h_{tw} \rightarrow h_{tw} = h_{hw} - h = 50$ m

[1 mark]

ii)

$$N \text{ (in rpm)} = \frac{f \text{ (in Hz)} \times 60}{p/2}$$

$$\Rightarrow p = \frac{2 \times f \text{ (in Hz)} \times 60}{N \text{ (in rpm)}} = 75$$

[1 mark]

iii)

- Salient pole rotor construction is necessary to accommodate a high number of poles
- This is different from the cylindrical or round rotor construction used for thermal generators which operate a high speed (1500 or 3000 rpm) requiring only 2 or 4 poles

[1 mark each]

iv)

$P_{\text{in}} = \rho g Q h$
$P_{\text{out}} = N_g \times P_g$

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{N_g \times P_g}{\rho g Q h} \rightarrow Q = \frac{N_g \times P_g}{\eta \times \rho g h} = 18.47 \times 10^3 \text{ m}^3/\text{s}$$

[1 mark]

Offshore Transmission

b)

i)

- Charging current drawn by the cable capacitance limits the active power carrying capacity of a long distance AC cable
- With HVDC, charging current is neglible compared to AC allowing the entire current carrying capacity of the cable to be used for active power transmission for longer distances

[1 mark each - 2 marks]

Model Answers

ii)

- An offshore wind farm could be a weak AC system. Operation of LCC-HVDC is much more reliant on a strong AC system than operation of a VSC-HVDC.

- An LCC-HVDC converter station has a much bigger footprint than an equivalent VSC-HVDC converter station due to higher reactive power and filtering requirement and special transformers needed for the former. A compact VSC-HVDC converter station suits the offshore platform.

- Stronger and lighter polymeric cables can be used for VSC-HVDC as there is no reversal of direct voltage polarity but can not be used for LCC-HVDC where power flow is reversed by changing the direct voltage polarity.

[1 mark each – 3 marks]

Model Answers